RV Institute of Technology and Management

Department of Master of Computer Applications

VIVA VOCE: QUESTIONS AND ANSWERS

AI Emergency Response Platform

How to use this. Answers are short so you can say them aloud. Learn the idea, not the exact words. Be honest about status: the project is in the **design stage**; the AI model is **not yet selected**, and nothing is implemented yet. Say “planned” or “proposed” for features. If you do not know something, say what you would check, not a guess.

A. Motivation, problem and objectives

Q1. Why did you choose this project?

Reporting an emergency by call or message often has unclear location, inconsistent descriptions and no feedback to the reporter. I wanted one web platform that makes reporting fast, helps sort reports using AI, and lets the reporter see progress.

Q2. What problem does it solve?

Four things: incomplete location information, inconsistent descriptions, slow manual categorisation, and no visibility of progress for the reporter. The platform reports, classifies, routes and tracks in one place.

Q3. What are your main objectives?

An easy report form (description, optional image, GPS, map or typed location); AI-suggested category and priority; routing to the right responder group; status tracking and notifications; an admin dashboard with analytics; secure login with roles.

Q4. What is the scope and what is outside it?

It is a college demonstration. AI only recommends and a human decides. It does not contact police, fire or ambulance services and does not replace official emergency numbers. Email, SMS and push alerts are future work.

Q5. Who are the users?

Three roles: User (reports and tracks), Responder (reviews, accepts, updates, resolves) and Admin (assigns, corrects AI results, manages users, views analytics).

B. Existing systems and literature survey

Q6. Which existing systems or papers did you study?

Seven papers. Main ones: Ignaco (2021), an Android incident reporting app with GPS, photo, responder and status tracking; Peralta Ascue et al. (2026), a reporting app with category routing and dashboard; Yoon et al. (2025), machine learning on 911 incident text; Sukhwani and Shaw (2020), disaster apps in India; Domfeh and Dancy (2025), human-AI decisions in disasters.

Q7. What do existing systems already provide?

Reporting with description, photo and GPS, responder assignment, status tracking and admin statistics. So these features are not new by themselves.

Q8. What is the research gap?

In the papers I reviewed, the reporting apps do not describe AI-based category or priority, and two list AI as future work. ML classification of emergency text is studied separately on existing records. My project integrates reporting, AI recommendation, routing and tracking in one platform.

Q9. Are you the first to do this?

No, I do not claim that. Other systems I did not review may have some of these features. My contribution is the integration, and I plan to extend the survey.

Q10. What did Yoon et al. find, and why does it matter to you?

They classified 3,564 fire-department incident descriptions into nine categories using TF-IDF with Naive Bayes, Random Forest and SVM. Random Forest reached 93% accuracy. It shows text classification of incidents is feasible with simple models, and gives me the evaluation metrics to reuse.

Q11. Why web application when most studied systems are mobile apps?

A web app works on any device with a browser, needs no installation and is easy to demonstrate. It can be made mobile-friendly. A mobile app is a possible future enhancement.

C. Technology: React, Node.js, Express, MongoDB

Q12. Why React?

React builds the interface from reusable components and updates only the parts that change. Hooks such as useState and useEffect manage state and data loading. It suits dashboards and forms with many repeated parts.

Q13. What is Node.js and why use it?

Node.js runs JavaScript on the server, so I use one language on frontend and backend. It handles many requests efficiently and has a large package ecosystem.

Q14. What is Express.js and what is middleware?

Express is a small web framework for Node that handles routes. Middleware is a function that runs before the route handler, with access to the request, response and a next function. I plan to use it for authentication, role checks, validation and error handling.

Q15. Why MongoDB instead of MySQL?

Incident records have optional and varying fields (image, coordinates, AI fields), which fit flexible documents. MongoDB also supports location queries. MySQL would also work; the choice is a design decision, not the only possibility.

Q16. What is Mongoose?

An object-data mapping library for MongoDB. It lets me define schemas with types, required fields and allowed values, and validates data before saving.

Q17. What is a REST API?

An API that uses standard HTTP methods on resources: GET reads, POST creates, PUT updates, DELETE removes. It is stateless and returns JSON with status codes such as 200, 400, 401, 403, 404.

D. Security: JWT, hashing, roles

Q18. What is JWT and how does login work?

A JSON Web Token has three parts: header, payload and signature. After a correct login the server signs a token containing the user id and role with a secret key. The frontend sends it in the Authorization header on every request, and the server verifies it.

Q19. Why hash passwords and how?

Plain passwords must never be stored. A hashing library such as bcrypt creates a salted one-way hash; at login the entered password is hashed and compared. Even if the database leaks, passwords are not readable.

Q20. What is role-based access control?

Each user has a role and each route allows only certain roles. The backend checks the role after verifying the token. Hiding a button in React is not enough because someone can call the API directly.

Q21. How do you stop a user from reading another user's report?

An ownership check in the controller: a User can open only incidents where userId equals their id. Responders see only their group or assigned incidents. Otherwise the server returns 403.

Q22. What other security measures are planned?

Server-side validation, image type and size checks, secrets in environment variables, CORS limited to the frontend, output encoding against XSS, and Mongoose queries built from validated values.

E. AI component

Q23. What is the role of AI here?

It reads the report and recommends a category and priority with a short summary and reason. It does not decide anything: a human responder or admin verifies, and the AI never contacts emergency services.

Q24. Which AI model or algorithm will you use and why?

Not finalised yet. I will compare three options: a rule-based baseline, TF-IDF with a classic ML classifier, and a pretrained language-model API. I will choose after testing on sample reports, considering accuracy, cost, ease of explaining, privacy and data availability. A rule-based baseline can also serve as a fallback.

Q25. How will classification and priority work?

Input is the description (and optional image and location text). The backend sends it to the AI and asks for JSON with category, priority, summary and reason. The backend checks that the category and priority are in the allowed lists before saving.

Q26. What if the AI is wrong or unavailable?

The report is saved before the AI runs, so it is never lost. If the AI fails or gives invalid output, aiStatus is set to failed, the report stays Submitted and the Admin is notified for manual review. If the AI is wrong, a responder or admin corrects it; I keep the AI value and the corrected value separately.

Q27. Will you show a confidence percentage?

Only if the chosen model gives a meaningful, calibrated value. I will not show an invented percentage. For example, a language model's own stated confidence is not reliable.

Q28. Can a user trick the AI by writing instructions in the report?

That is a known risk called prompt injection. Because the AI only returns a recommendation and the backend validates it and controls all database actions, the AI cannot change permissions or data directly.

Q29. How will you evaluate the AI?

I will build a small labelled set of sample reports and measure accuracy, precision, recall, F1 and a confusion matrix. For priority, recall for High and Critical matters most because missing a serious case is worse than over-flagging. I have no results yet.

F. Design: HLD, LLD, database

Q30. What is the difference between HLD and LLD?

HLD shows the main components and how they connect, for example frontend, API, AI adapter and database. LLD defines the inside details: fields, routes, validation, status transitions and error codes.

Q31. Explain your architecture.

A React frontend calls an Express REST API with a JWT. The API runs auth and validation middleware and four services: incident management, AI adapter, routing, and notifications with history. Data goes to MongoDB and image storage; the AI model is called only from the backend.

Q32. Why is the AI behind an adapter?

So the model can be changed without touching other modules, the API key stays on the server, and every AI answer is validated before use. There is no public AI route.

Q33. Explain your database collections.

Four collections: User, Incident, IncidentHistory and Notification. Incident refers to the reporter and assigned responder by id. History stores every status change with who, when and a note.

Q34. What is the status flow?

Submitted, AI Analyzed, Assigned, Accepted, In Progress, Resolved, plus Pending Assignment when no responder is available. Only valid transitions are allowed; for example Submitted cannot jump to Resolved.

Q35. What if two responders accept the same incident?

The accept action uses an atomic update that succeeds only if the incident is still unaccepted. The first wins and the second receives a conflict message (409).

G. Location, image and features

Q36. How will GPS and map location work?

The browser geolocation API gives latitude and longitude after the user allows permission. The user can instead click a point on a map or type a description. If permission is denied, the form explains and the other options remain.

Q37. How will image upload work?

One optional JPG, JPEG or PNG file, checked for type and size (proposed limit 5 MB), renamed safely and stored; the database keeps only the path. A file-upload middleware will handle it.

H. Status, testing, limitations, future

Q38. What have you completed so far?

Problem definition, literature survey, SRS, HLD and LLD. Implementation has not started; that is the next stage. I will not claim anything as implemented until it works.

Q39. How will you test the system?

Unit tests for functions, API tests for routes, integration tests for the full flow, role and permission tests, form validation tests, and error cases such as GPS denied, AI timeout and invalid image. AI evaluation is separate from software testing.

Q40. What are the limitations?

AI can be wrong, so a human must verify. Responders are internal demo accounts. Notifications are in-app only. It needs internet. Image analysis depends on the chosen model.

Q41. What is the future scope?

Email, SMS and push notifications, a mobile app, better image analysis, predicting incident hotspots, and integration with real emergency services only with proper authorisation.

Q42. Why not just call the emergency number?

The platform does not replace official numbers. It helps with structured reporting, location and tracking inside an organisation or campus, and in a real emergency users should still contact official services.

Tip for all answers: start with a one-line answer, add one reason, then one example from your project.